

RECENT DEVELOPMENTS

Technology	Developments
Bluetooth	Recent developments include BLE with 24Mbps data transfer speeds. With mesh networking still under wraps, it is expected to be a major change in the coming years.
Wi-Fi	A new standard, 802.11ad, with theoretical speeds of 7Gbps would remove any need for wires at all. With support for wireless devices and Internet connectivity, this is expected to be a big step in development.
NFC	Low-energy chips with easy-to-implement modules will see a major rise in customer use cases. The ease of functioning is already an attractive feature with the users. Security would further help in making a significant change.
ZigBee	Mesh networking and ease of implementation have been the driving factors behind ZigBee. Large-scale implementation in production lines and Industrial IoT is expected to further boost the technology.

use, and so it has found uses in security applications and bank transfers. T. Anand, managing director, Knewron, agrees, "Even if you are near the payment kiosk and the NFC tag is in your pocket, then payment is not happening by mistake. When you employ NFC, you basically touch the interface to the tag, making the distance almost zero."

ZigBee. "ZigBee has wide uses in the home automation space, since it is a mature technology supporting mesh and large networks," says Siddharth Unny, TL Micro Automation Pvt Ltd. He adds, "There was some talk of ZigBee being included in smartphones, enabling these to directly communicate with ZigBee networks. However, with the introduction of mesh networking in Bluetooth, addition of ZigBee to smartphones seems unlikely."

Wireless fidelity. With the incorporation of Wi-Fi into mobile handsets (smartphones), it has become a common household name, and is often wrongly understood for the Internet. "Wi-Fi chipsets have been steadily improving in all respects, supporting greater speed, easier connectivity and low power operation," adds Unny.

With regular advancements, newer and updated versions of communication chips are being launched on a daily basis. Let us take a look at the recent

developing trends.

Personal networking connections. Unny says, "An exciting new development is Bluetooth mesh networking, which will allow Internet of Things (IoT) devices to be connected using Bluetooth, over large networks."

ZigBee modules also provide a system to connect a mesh network. These often provide dedicated software, which provides a user-friendly graphical interface in configuring the modules. "Serial data monitoring with data logging, wireless range test applications, framing data packets, discovering neighbouring ZigBee modules and forming mesh networks with drag-and-drop configuration are some additional features," adds Kaustubh Kamataki, chief engineer, FluxGen Engineering Technologies.

The high-speed capability of Wi-Fi allows gaming among a group of friends or accessing a private server as some of the non-Internet uses of Wi-Fi. With ever-developing protocols it can also be used for the setup of a private Intranet system, accessible only to specified users.

Low power and cost, high applicability. We have Bluetooth Low Energy (BLE) connectivity modules, with battery consumption in milli-amperes that find heavy use in IoT applications. XR and ZR modules from Atmel integrate a low-power Bluetooth system on

chip (SoC) with an Atmel SMART ARM Cortex-M0 microcontroller (MCU) and a Bluetooth transceiver.

"We are excited to bring additional BLE solutions to the rapidly-growing wearable and IoT markets. Catering to wireless application designers of all levels, these give designers the critical tools to quickly and easily integrate Bluetooth Smart systems and connectivity," says Bert Fransis, senior product marketing manager - Bluetooth Solutions, Atmel Corp.

CC3200 SimpleLink Wi-Fi and IoT solution, a single-chip wireless MCU, is another Wi-Fi module from Texas Instruments. This well-established module consists of an integrated MCU, reducing the need of peripherals. ARM core based Nordic nRF51822 and 8051 based TC CC2540 are other options.

Such products allow designers with no/less prior Bluetooth or RF design experience to add BLE connectivity to their designs. The new modules add to the existing BLE solutions, giving designers additional options to select from, cost-effective, ultra-small modules that require an external antenna or from fully FCC/CE/IC-certified modules with integrated antennae.

Comparing these BLE chips to the time when we carefully switched off Bluetooth after every use seems overkill.

Multiple applications in a small form factor. A major trend has been the integration of multiple technologies into a single chip. Anand agrees, "Today we are merging at least two technologies together like BLE + RF or BLE + Wi-Fi." Multiple chipsets on a single chip reduce integration of several different components and also bill of materials.

LILY-W1 is a single-band 802.11n 2.4GHz Wi-Fi transceiver module from U-Blox. It provides Wi-Fi connectivity to a very broad range of products, along